Kashmir Valley Transit
Outputs Explained — Reader's Guide

Every artefact the engine emits, who should open it, which section to focus on, and how it ties back to the briefing-deck numbers. Companion to Kashmir_Transit_Headway_Fleet_Fundamentals.pdf (which covers the maths).

Engine v3.3.5 · May 2026

For Principal Secretary, RTOs, and the operations team

1. The artefact map

Every engine run produces the same eight artefacts in the **outputs_v3.3.5/** directory. Different audiences open different files; pick the row that matches you.

Artefact	File	Best read by
RTO Master Workbook	Kashmir_Route_Frequency_Plan_v3.3.5_RTO.xlsx	RTOs, Principal Secretary, IAS reviewers
Legacy 4-sheet Workbook	Kashmir_Route_Frequency_Plan_v3.xlsx	Engineering team, internal review
Master Transit Map (HTML)	Master_Transit_Map_Kashmir_v3.html	Anyone wanting an interactive overview
Per-route maps	route_maps_kashmir/<ID>.html (one per active route)	RTO drilling into a single corridor
Operational CSV	Rationalised_Routes_Kashmir_v3.csv	Data team, dashboard, statisticians
GeoJSON network	Rationalised_Routes_Kashmir_v3.geojson	GIS team (load into QGIS / ArcGIS)
Rationalisation Log	Rationalisation_Log_Kashmir_v3.csv	Anyone questioning why a route was merged / upgra
Passenger Impact	Passenger_Impact_Kashmir_v3.csv	Communications team, public-facing summaries
Pipeline log	engine_run_v3.3.5.log	Debugging only — QC trace from the engine

2. The RTO Master Workbook — sheet by sheet

Kashmir_Route_Frequency_Plan_v3.3.5_RTO.xlsx is the primary deliverable for sign-off. Nine sheets, each with a specific reader.

Sheet	Purpose	Who reads it	Focus on
1. Cover & Sign-off	Title block, 8 KPI tiles, legend, signature lines.	Principal Secretary	KPI tiles + sign-off block at bottom.
2. Network Summary	Network composition, fleet composition, headway distribution.	RTO, IAS	Headway distribution table + total reconciliati…
3. Route-Level Plan	All 342 routes with action, headway, fleet, load flag, map link.	RTOs (working sheet)	Filter on Action_Taken or Priority_Band. Click…
4. Operator Absorption	Permits merged into trunks, broken down by operator class with staffing/buyback estimates.	Finance, Welfare Assn liaison	■12.96 Cr buyback estimate … ask (TOTAL row…
5. Trunk Detail	All 50 UPGRADED_TO_TRUNK + HP routes only.	Backbone reviewers	Fleet vs headway vs SSCL_ID — verify SSC…
6. Social Obligation	All Social_Flag routes (KP townships, hospitals, women's colleges).	Political clearance reviewers	Verify nothing critical demoted to LP.
7. Tourist & Seasonal	Tourist_Corridor + Seasonal_Operability routes.	JKTDC, JKRTC seasonal planners	Winter-suspended list (Mughal Road, Sinthan…
8. Calibration & Sources	Headway targets, CHALO calibration, data sources.	Anyone questioning the numbers.	Mode-share assumption + capture-scale deri…
9. Limitations	Known gaps (Phase-2 backlog).	IAS — must see before sign-off.	All bullets — these are caveats that need to r…

How to read Sheet 3 (Route-Level Plan) — column groups

The 28 columns are organised into five visual groups by colour:

Group (column-header colour)	What it tells you
Identity (navy)	Route_Code, Old_Route_ID, New_Route_ID, Route_Name, Action_Taken, Priority_Band. Fr…
Service (teal)	Route_KM, Cycle_Time_Min, Headway_Min, Fleet_Required, HPV/MPV/LPV counts, Bus_Ty…
Demand (purple)	Population_Served, Daily_Demand_Pax, Daily_Capacity_Pax, Load_Flag, Subsidy_Risk_Fla…
Impact (red)	Displaced_Operator_Class, Num_Permits_Affected, Recommended_Action, RTO_Remarks (…
Audit (amber)	Final_CDI, Map_Link. Why this route got its band, plus a hyperlink to its individual map.

Sheet 3 is set up for landscape A3 printing. Headers repeat on each page, auto-filter is on, and the identity block (cols A–F) freezes when you scroll right.

3. Rationalised_Routes_Kashmir_v3.csv — column by column

The single most useful file for anyone working with the data downstream. Every column the engine writes; every column the dashboard reads.

Column	Meaning
Route_Code	12-char tehsil+sector+stop code — or TMP-K#### placeholder while the stops-master is pending
Route_ID	Source permit ID from existing-routes.csv (R0001…R0342)
Route_Name	ORIGIN ↔ DESTINATION (engine reconstructs from Origin/Destination when blank)
Action_Taken	UPGRADED_TO_TRUNK · RETAINED_AS_FEEDER · MERGED_INTO_TRUNK
New_Route_ID	TRK-NNN for trunks, FDR-NNN for feeders, blank for merged
Displaced_Operator_Class	Private Minibus / LPV / Tempo / HPV Bus / JKRTC. Only for MERGED rows.
Route_KM	Route length from OSRM (or circuity fallback)
Route_Type	Urban · Peri_Urban · Regional_District (typology)
OSRM_Duration_S	Free-flow OSRM travel time in seconds, no congestion
Cycle_Time_Min	Round-trip cycle including congestion, dwells, junctions, bridge bottleneck
Congestion_Zone	City_Core · Peri_Urban · Highway (drives the congestion multiplier)
N_Stops_Estimated	Virtual stops counted at 500m spacing
Stop_Penalty_Min	Total dwell penalty on the route
Sharp_Turns	Junction count from OSRM geometry (each adds 30s)
Junction_Penalty_Min	Total junction-time penalty
Pop_Score	Normalised catchment population (0–1)
POI_Score	Normalised POI gravity score (0–1)
Road_Multiplier	Tie-breaker for routes near a Jenks band boundary
Final_CDI	= Pop_Score × 0.5 + POI_Score × 0.5 → input to Jenks bands
Social_Flag	True for KP townships / hospitals / women's colleges (LP→MP floor)
Priority_Band	HP / MP / LP (high / medium / low)
Headway_Min	Recommended target headway
Fleet_Required	Recommended bus count = CEIL(Cycle/Headway × 1.15)
HPV_Count	12-metre buses on this route
MPV_Count	9-metre buses on this route
LPV_Count	Tempo / minibus on this route (feeder LPV-category only)
CMP_Trunk	True for the 45 permits matched to the 30 SSCL CHALO routes

CMP_Route_ID	Matched SSCL ID (SSCL-01…SSCL-30) for CMP_Trunk routes
Population_Served	Apportioned (dedup-residual) catchment count for the route
Population_Served_Raw	Pre-apportionment buffer count (used in Phase-4 demand)
Corridor_Competitors	# overlapping route buffers in the corridor
HV_POI_Count	High-value POI count in the route's 250m buffer
Overlap_Metric	Mean overlap with peer routes (0–1)
Geo_Source	OSRM · OSRM+JhelumBridge · Circuity · Circuity-River
Tourist_Corridor	True if route passes near a tourist zone (geom or endpoint test)
Seasonal_Operability	Year_Round · Seasonal · Winter_Suspended
District_HQ_Floor	True if route touches a district HQ (LP→MP floor)
SSCL_CDI_Conflict	True for non-SSCL routes with CDI ≥ worst SSCL trunk + 0.2 (planner review)
Daily_Trips	Service-day one-way trips offered at the target headway
Daily_KM	Service-day scheduled KM = Daily_Trips × Route_KM
Daily_Capacity_Pax	Daily seats offered = Fleet × Capacity × (960 / Cycle_Time_Min)
Daily_Demand_Pax	Estimated daily riders (Phase-4 corridor-share model)
Load_Ratio	Daily_Demand_Pax ÷ Daily_Capacity_Pax
Load_Flag	Diagnostic: Green / Amber_Under / Amber_Tight / Red_Overload / Red_NoCapacity
Pax_Journey_Time_Min	Estimated total passenger journey time (access + wait + ride)
Journey_Time_Flag	True if Pax_Journey_Time exceeds the 45-min comfort threshold
Daily_Revenue_INR	Daily fare-recovery estimate at ■10/trip
Daily_Op_Cost_INR	Daily operating cost at ■65/km
Viability_Ratio	Daily_Revenue ÷ Daily_Op_Cost (1.0 = self-funding)
Subsidy_Risk_Flag	True if Viability_Ratio < 0.60 AND not Social_Flag
Emissions_GCO2_Daily	Daily emissions estimate (lower for SSCL e-bus routes)
Equity_Score	Composite equity score normalised across the network
Map_File	Relative path to the route's individual HTML map

4. Master & per-route HTML maps

Master_Transit_Map_Kashmir_v3.html is a Folium-rendered interactive map. Opens in any modern browser — no GIS skills required. Use it for:

- **Overview presentations** — toggle the trunk / feeder / SSCL / regional layers in the right sidebar.
- **Selecting a route** — click any line; the popup shows the route's headway, fleet, KPIs and a link to the individual map.
- **POI overlay** — turn on the POI layer to see why a route scored highly on its POI_Score.
- **Winter / summer comparison** — re-run the engine with WINTER_SCENARIO=True to generate the winter version of the map.

Map tiles are OpenStreetMap / CartoDB. They do not represent the official J&K; political boundary — a disclaimer is shown in the map's footer.

Per-route maps under route_maps_kashmir/

One HTML file per active route, named after its New_Route_ID (e.g. **TRK-001.html**, **FDR-047.html**, **SSCL-15.html**). Each shows just that route's geometry, its catchment polygon, the POIs it touches, and a popup with all key KPIs. Linked from the Master map and from the RTO workbook's Sheet 3.

5. GeoJSON for GIS teams

Rationalised_Routes_Kashmir_v3.geojson contains the 207 active routes as LineString features. Load into QGIS or ArcGIS for spatial joins, road-segment analysis, or overlay against your own datasets (ward boundaries, depot locations, accident hotspots, etc.). Every property field is the same as the CSV column.

6. Rationalisation Log — defending decisions

Rationalisation_Log_Kashmir_v3.csv is identical to the main routes CSV except for one extra column at the end: **Reasoning_String**. This is the human-readable explanation of why each route got its band, action, fleet, and headway.

Use this when an operator or RTO official questions a specific decision. Example reasoning string for a merged route:

```
absorbs this permit (overlap 0.78 with parallel trunk; CDI 0.51 vs neighbour 0.78). Distinct catchment: 262 reside
```

7. Passenger Impact CSV — the public-facing summary

Passenger_Impact_Kashmir_v3.csv is the simplified, passenger-facing view. Only active routes, only the columns a citizen reading a notice would care about.

Column	What it tells the passenger
New_Route_ID	The route number you'll see on the bus
Route_Name	Origin ↔ Destination
Action_Taken	Trunk / Feeder / Merged
Route_Type	Urban / Peri-urban / Regional
Priority_Band	HP / MP / LP
Headway_Min	Wait time between buses
Fleet_Required	Number of buses on this route
HPV_Count	Large buses
MPV_Count	Medium buses
LPV_Count	Small minibuses / tempos
CMP_Trunk	Yes/No — is this an SSCL e-bus route?
Population_Served	Residents in the 400m walking buffer
Social_Flag	Yes/No — is this a protected lifeline?

Use this CSV as the source for citizen-facing materials: notice-board posters, app schedules, the website route list, and the bus-stop printed timetable.

8. The pipeline log — when something looks off

engine_run_v3.3.5.log contains the full step-by-step trace of what the engine did. Don't open it day-to-day — it's diagnostic. Open it when:

- A QC check fails — the log will name the check and the offending routes.
- Calibration drifts more than ±15% from CHALO — check the population fallback and capture-scale lines.
- Outputs don't match what the dashboard shows — the log is the source of truth, the CSV is the export.

Key log lines and what they mean

Log line	Means

Tourist boost applied: 1.30x on N routes	Population multiplier applied to N tourist routes
Apportioned Population_Served. New naive sum: X	Population deduplication ran; X is the sum after sharing
Priority bands — HP: N · MP: N · LP: N	Jenks classification result
Load_Flag: Green=N Amber_Under=N Amber_Tight=N Red_Overload=N	Phase-4 diagnostic spread
✓ ALL QC CHECKS PASSED — workbook ready for export	Engine cleared all 8 quality checks — outputs are safe to ship
✗ rasterstats not importable / WorldPop raster not found	Population source missing — engine will refuse to ship unless

9. Tying outputs back to the briefing decks

Every number on every slide of the briefing decks traces back to one of these files. Quick cross-reference for an IAS reviewer asking 'where does that number come from?'

Slide / number	Source
207 active routes	Routes CSV — COUNT WHERE Action_Taken ≠ MERGED_INTO_TRUNK
988 total fleet (138/730/120)	Routes CSV — SUM(Fleet_Required) on active rows; HPV+MPV+LPV cols
1,158,399 residents covered (69.78%)	engine_run log line 'Deduplicated network population'
45 SSCL trunks / 30 matched	Routes CSV — COUNT WHERE CMP_Trunk=True
69 tourist corridors	Routes CSV — COUNT WHERE Tourist_Corridor=True
12.96 Cr buyback ask	RTO Workbook · Sheet 4 · TOTAL row, F column (SUM formula)
+9.7% per-route calibration	cross_eval_v3.3.5.log — objective section
8/8 QC checks	engine_run log — '✓ ALL QC CHECKS PASSED' line
0 Red_Overload routes	Routes CSV — COUNT WHERE Load_Flag='Red_Overload'

Reading order for a brand-new reviewer

1. Open the diagrammatic pitch deck — Kashmir_Transit_Diagrammatic_Pitch.pptx.
2. Open the RTO Workbook · Sheet 1 (Cover) for the headline numbers.
3. Open the Master HTML map and click 5–10 random routes to feel the geography.
4. If you want to defend a number — find it here in this guide and follow the trace.
5. If the maths bothers you — open Kashmir_Transit_Headway_Fleet_Fundamentals.pdf instead. That covers the formulas; this guide covers the artefacts.